

DATA SCIENCE PROJECT REPORT

Marketing Campaign Analysis

Customer Analytics, Campaign Intelligence and Targeting

Python | MySQL | Microsoft Power BI

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Project type	Exploratory Data Analysis and Business Intelligence
Dataset	56,000 customer records
Repository	github.com/Ameena-Juhi-99/Marketing-Campaign-Analysis-PowerBI

Executive Summary

This project delivers an end-to-end analysis of marketing campaign and customer behaviour using Python, MySQL and Power BI. The analytical workflow converts a 56,000-record source dataset into a cleaned 58-field customer table, reusable SQL assets and a four-page interactive dashboard.

Core result: The latest campaign records a 14.76% response rate. Premium-value customers respond at 27.61%, making customer value segmentation central to campaign prioritisation.

Project Objectives

- Measure overall and campaign-level acceptance and response.
- Understand spending patterns across products, countries and income bands.
- Identify the strongest customer segments and purchase channels.
- Detect under-served customer groups and define ideal target profiles.
- Present validated findings through an interactive Power BI report.
- Translate analysis into actionable marketing recommendations.

Headline KPIs

Metric	Result
Total customers	56,000
Total spending	35.86M
Total purchases	619.7K
Latest response rate	14.76%
Average customer spending	640.33
Complaint rate	0.76%
Ideal targets	4,504
Under-served customers	16,765

1. Business Problem and Analytical Questions

Marketing teams need to understand not only whether a campaign succeeded, but also which customers responded, what they purchased, where they purchased it and which groups remain insufficiently served. The project addresses this need through a reproducible analytical workflow and an interactive decision-support dashboard.

Business Questions

1. What is the overall response rate, and which campaigns perform best?
2. Which products and purchase channels generate the greatest activity?
3. How do response and spending vary by country and customer segment?
4. Which demographic and behavioural features describe ideal targets?
5. Which customers appear under-served?
6. How does website engagement differ between responders and non-responders?

2. Dataset Description

Dataset asset	Purpose
marketing_campaign_data.csv	Original 56,000-row customer campaign dataset with 28 source fields.
marketing_data_dictionary.csv	Definitions of the original fields.
marketing_campaign_cleaned.csv	Cleaned and engineered 56,000-row, 58-field analytical dataset.
Power BI fact CSVs	Campaign, product and channel tables used with the central customer table.

The source data covers demographics, income, household composition, product spending, purchase channels, promotions, campaign acceptance, complaints, country and website engagement.

3. Methodology

The implementation follows a transparent sequence so that each dashboard measure can be traced back to a source field, engineered feature or analytical query.

Step	Stage	Output
1	Load	Read the campaign dataset and field dictionary.
2	Audit	Check shape, types, nulls, duplicates, categories and ranges.
3	Clean	Standardise fields and validate identifiers and numeric values.
4	Engineer	Create customer, value, engagement, campaign and targeting features.
5	Analyse	Perform univariate, bivariate and segment-level EDA in Python.
6	Store	Load the analytical table into MySQL and run reconciliation checks.
7	Report	Model the data, create DAX measures and build four Power BI pages.
8	Validate	Test relationships, slicers, reset actions, navigation and refresh.

Data Quality Controls

- Confirmed zero missing customer IDs, income values and total-spending values after preparation.
- Verified one unique central record per customer ID.
- Validated that cleaned, MySQL and Power BI customer totals reconcile to 56,000.
- Checked response, spending, purchase and complaint measures between analytical layers.
- Preserved transparent, rule-based definitions for ideal-target and under-served segments.

4. Data Cleaning and Feature Engineering

Feature group	Representative fields
Customer profile	Age, Age_Group, Education_Group, Relationship_Status, Family_Customer
Household	Total_Children, Has_Children
Customer value	Total_Spend, Total_Purchases, Average_Spend_Per_Purchase, Customer_Value_Segment
Engagement	Customer_Tenure_Days, Customer_Tenure_Months, High_Web_Engagement, Preferred_Channel
Campaign	Historical_Accepted_Campaigns, Total_Accepted_Campaigns, Campaign_Responder
Targeting	Ideal_Target_Customer, Under_Served_Customer, Income_Band, Income_Outlier_Flag

5. Python Exploratory Data Analysis

The Jupyter Notebook documents data loading, dictionary inspection, data-quality checks, cleaning, feature engineering, exploratory analysis, customer segmentation, targeting logic, recommendations and the export of Power BI-ready data.

- Campaign acceptance comparisons across all six campaigns.
- Product spending and purchase-channel distributions.
- Country, age, income, education and household comparisons.
- Customer value and response segmentation.
- Website visits, deal behaviour and complaint analysis.
- Ideal-target and under-served customer profiling.

Reproducibility: The notebook runs from beginning to end without cell errors and exports the cleaned customer table plus the three Power BI fact tables.

6. MySQL Implementation

MySQL provides the structured analytical layer. Seven numbered scripts separate schema creation, data loading and verification, KPI analysis, segmentation, product and channel analysis, advanced SQL and reporting views.

Script	Main responsibility
01_database_and_schema_setup.sql	Database and 58-column customer schema with primary key and validation constraints.
02_data_load_and_verification.sql	Reproducible load instructions and reconciliation queries.
03_executive_kpi_queries.sql	Executive totals and campaign-performance analysis.
04_demographic_segmentation_queries.sql	Age, income, education, relationship and country segmentation.
05_product_and_channel_queries.sql	Product, purchase channel, promotion and deal analysis.
06_advanced_window_queries.sql	Analytical ranking and window-function examples.
07_reporting_views.sql	Reusable views for reporting and dashboard support.

SQL Concepts Demonstrated

- DDL and data loading
- Primary-key and check constraints
- Aggregate functions and grouped analysis
- CASE expressions and CTEs
- UNION ALL
- Window functions and ranking
- Reusable reporting views

Verification baseline: 56,000 customer rows, no missing IDs, no missing income and no missing total-spending values.

7. Power BI Data Model and Measures

The report uses a star-style model centred on the Customers table. Campaign, product and channel fact tables connect through customer ID, allowing the demographic slicers to filter every report page consistently.

Table	Role	Relationship
Customers	Central dimension; one row per customer	One side
Campaign_Performance	Customer-campaign acceptance records	Many-to-one through ID
Product_Spending	Customer-product spending records	Many-to-one through ID
Channel_Purchases	Customer-channel purchase records	Many-to-one through ID
KPI_Measures	Reusable DAX measures	Measure holder

Interaction Design

- Five synchronized dropdown slicers: Country, Education, Marital Status, Age Group and Income Band.
- Reset Filters button on each page.
- Bottom page navigator with active-page highlighting.
- Active many-to-one relationships with single-direction filtering.
- Consistent KPI cards, colours, titles, labels and chart containers across all pages.

8. Dashboard Page 1 - Executive Overview

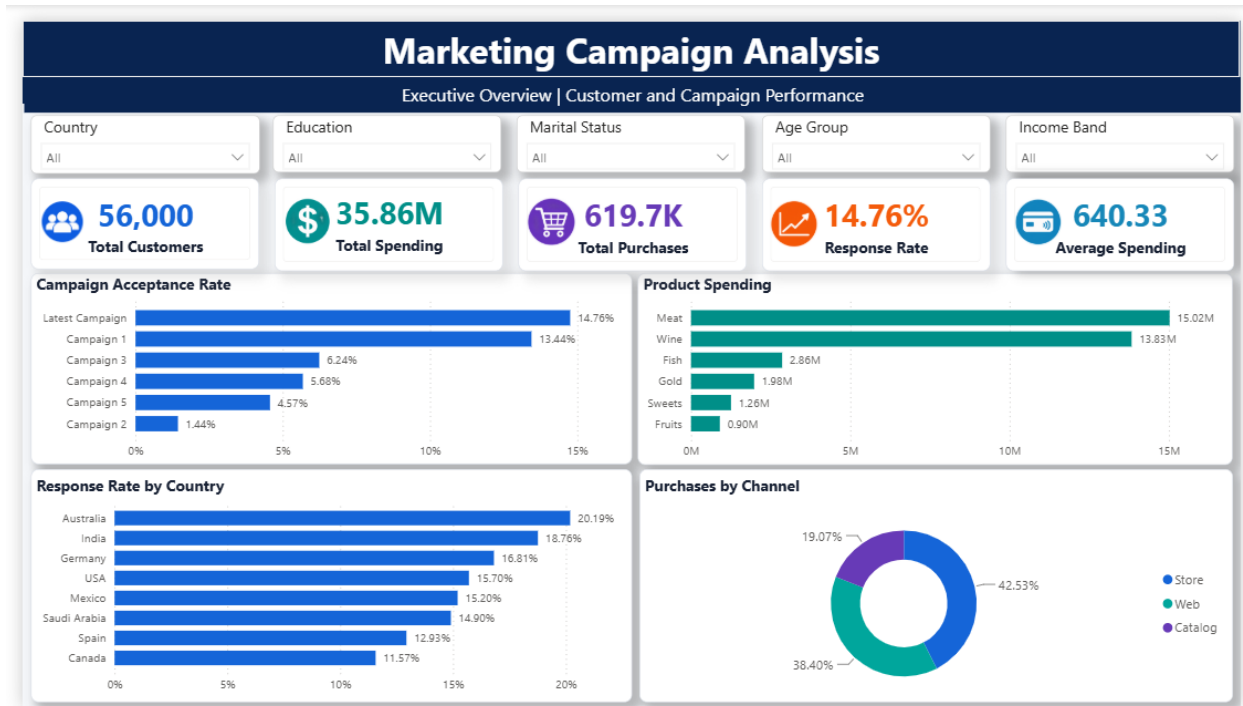


Figure: Executive Overview - Executive KPIs and high-level campaign, product, country and channel performance.

Key Takeaways

- The latest campaign leads with a 14.76% response rate; Campaign 1 follows at 13.44%.
- Meat and wine dominate product spending at 15.02M and 13.83M.
- Australia has the strongest country response at 20.19%; Canada is lowest among the displayed countries at 11.57%.
- Store purchases contribute 42.53%, web 38.40% and catalog 19.07%.

9. Dashboard Page 2 - Customer Segmentation



Figure: Customer Segmentation - Demographic, income, relationship and customer-value segmentation.

Key Takeaways

- The 30-39 and 40-49 age groups contain the largest customer populations.
- Average spending rises from 194.19 in the Low income band to 1,228.29 in Premium.
- Other and Not Partnered relationship groups show higher response than Partnered customers.
- Standard Value represents 50.00% of customers, Growth Value 44.94% and Premium Value 5.06%.

10. Dashboard Page 3 - Campaign Intelligence



Figure: Campaign Intelligence - Campaign acceptance, value-segment response and purchase behaviour.

Key Takeaways

- The analysis records 8,265 latest-campaign responders and 25,834 campaign acceptances in total.
- Premium-value customers respond at 27.61%, compared with 20.84% for Growth and 7.99% for Standard Value.
- Premium-customer purchases are strongest in Store, followed by Web, Catalog and Deal-Assisted activity.
- Non-responders average 5.27 website visits versus 4.58 for responders, suggesting conversion friction rather than insufficient traffic.

11. Dashboard Page 4 - Customer Targeting

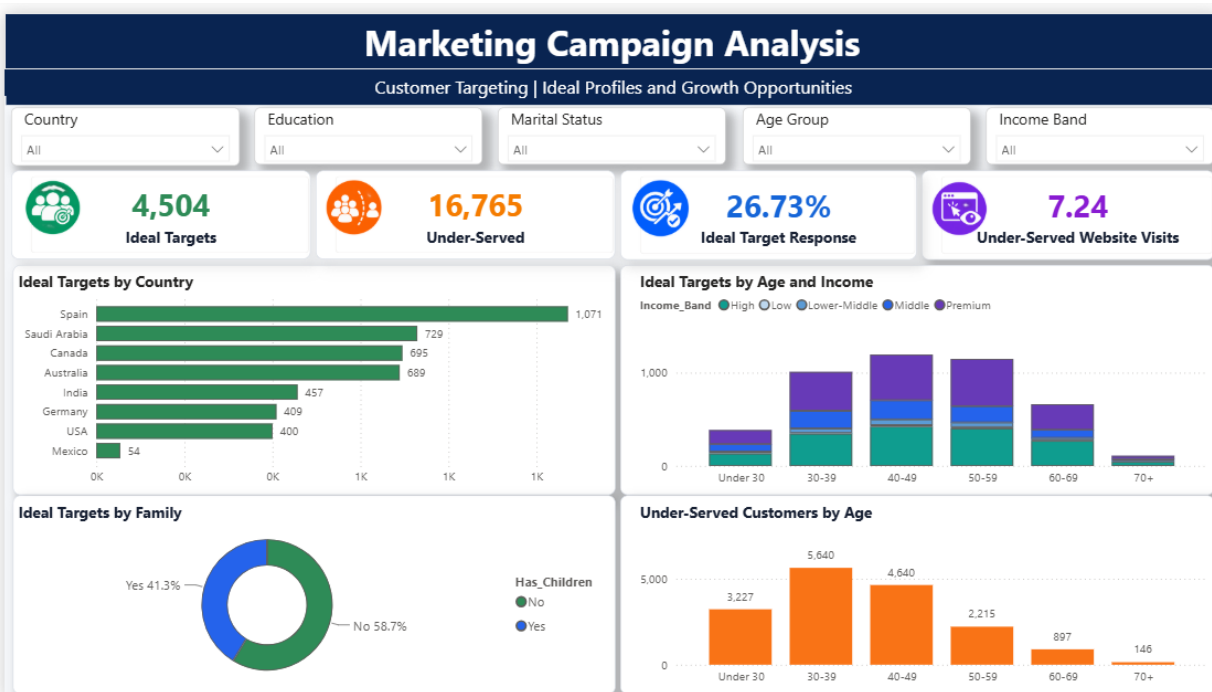


Figure: Customer Targeting - Ideal-target profiles and under-served growth opportunities.

Key Takeaways

- The ideal-target segment contains 4,504 customers and responds at 26.73%.
- Spain contains the most ideal targets, followed by Saudi Arabia, Canada and Australia.
- A majority of ideal targets have no children (58.7%); 41.3% have children.
- Under-served customers are concentrated in the 30-39 and 40-49 age groups.

12. Consolidated Findings

Area	Finding	Business implication
Campaigns	Latest and Campaign 1 outperform the remaining campaigns.	Replicate and test their audience and message strategy.
Customer value	Premium and Growth customers respond far more strongly than Standard Value.	Protect high-value segments and improve Standard conversion.
Products	Meat and wine account for most product spending.	Use them as anchors for bundles and cross-selling.
Channels	Store and web together dominate purchase activity.	Coordinate physical and digital journeys.
Countries	Response and ideal-target concentration vary meaningfully by country.	Localise targeting rather than using a uniform campaign.
Engagement	Non-responders visit the website more frequently.	Investigate landing-page, offer and checkout friction.
Opportunity	Under-served customers cluster in core working-age groups.	Prioritise age- and income-sensitive activation campaigns.

13. Business Recommendations

- Prioritise proven campaign designs. Reuse the targeting and messaging principles of the latest campaign and Campaign 1, then validate them through controlled A/B tests.
- Protect Premium and Growth customers. Use loyalty benefits, early access, tailored bundles and personalised cross-sell offers.
- Use channel-specific strategies. Retain stores for high-intent purchases, improve web conversion and use catalog selectively.
- Bundle high-performing products. Pair meat and wine with lower-spending categories to increase basket diversity.
- Activate under-served customers. Focus on the 30-49 age range while adapting campaigns by income, country and family profile.
- Improve web conversion quality. Review landing pages, offer relevance and checkout friction because higher traffic does not translate into higher response.

14. Validation and Testing

Test	Status	Evidence
Notebook execution	Passed	All cells run without errors.
Customer integrity	Passed	56,000 complete and unique customer IDs.
Database reconciliation	Passed	MySQL totals match the analytical baseline.
Model relationships	Passed	Active many-to-one relationships with correct filter direction.
Slicer behaviour	Passed	KPI cards and charts update across all report pages.
Slicer synchronization	Passed	All five slicers stay synchronized.
Reset and navigation	Passed	Reset buttons and page navigator work correctly.
Power BI refresh	Passed	Refresh completes without connection or file-path errors.

15. Limitations

- The dataset is a customer snapshot rather than a time-series campaign log.
- Campaign cost and attributable revenue are unavailable, so true return on investment cannot be calculated.
- Ideal-target and under-served classifications are transparent rule-based segments, not predictive-model outputs.
- Deal-assisted purchases overlap with channel behaviour and are not a mutually exclusive fourth purchase channel.

16. Conclusion

The project satisfies the core analytical requirement by combining reproducible Python preparation, structured MySQL analysis and an interactive four-page Power BI dashboard. It identifies where campaigns perform well, which customers create the greatest opportunity and how product, channel, demographic and engagement behaviour should influence future targeting.

Final outcome: A validated decision-support solution that converts 56,000 customer records into clear campaign, segmentation and targeting actions.